



Advanced Machine Learning

2020/2021

Exam R 12 of July 2021 Time: 2h00

Name:

Number:

Question #	Points	Your score
1	18	
2	12	
3	20	
4	25	
5	25	
	100	

Grader:

Pergunta 1 - Define 18 points

Define the following concepts in a clear and **rigorous** way. You may use a concrete example to **help you** explain the concept.

a) Vanishing Gradient.

A sua resposta:

b) Padding.

A sua resposta:

c) Theorem of Rosenblatt.

A sua resposta:

Pergunta 2 - Comment 12 points

Comment the following statements. **Remember**, to comment is not just to say agree/disagree for you have to justify your evaluation.

- a) If a feed forward neural network has linear activations functions for all neurons in the hidden layers, the quality of the result do not depend on the number of hidden layers.

Your answer:

Your answer:

- b) If your cost function is convex, gradient descent is guaranteed to find the optimal parameters for your model.

Your answer:

Your answer:

Pergunta 3 - Solve 20 points

Consider the decision tree of figure 1. Inside each node we show the number of representatives of each of two classes associated with that node. Compute the **information gain** associated with the root node. Moreover, what are possible values for X and Y so the parent node would have the **largest** possible information gain. Justify with rigor your answers.

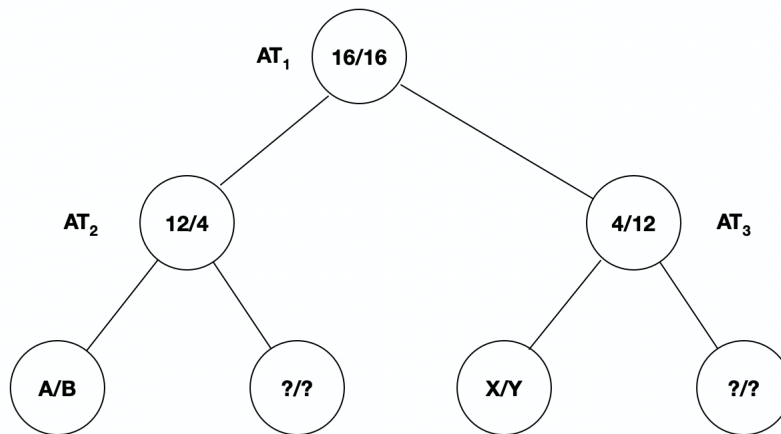


Figure 1: A decision tree

Your answer:

Your answer:

Pergunta 4 - [Explain/Describe](#) 25 points

Explain carefully what we need to do to your dataset **before** you apply a learning algorithm.

Your answer:

Your answer:

Pergunta 5 - [Architecture](#) 25 points

You are a world class detective and you want to find, among ten suspects – all UC students, which one is the murderer of a famous university professor. The only clue you have is **one handwritten character** left by the killer near the dead body. You followed a Coursera online course on Machine Learning and became an enthusiast of the subject and so you decide to apply a ML approach to replace you. Describe and justify carefully your solution.

Your answer:

Your answer: